A TWO-STAGE CLASSIFICATION OF LINEAR GENERALIZED STONE–WALES PATHS IN GOLDBERG–COXETER IMAGES OF THE DODECAHEDRON

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Abstract. We study literal linear generalized Stone–Wales (gSW) paths in the duals of $\mathrm{GC}(k,l)[C_{20}]$, where $k > l > 0$. Such a path has distinct vertices, an even number of vertices, degree-five endpoints, degree-six neighbours of its endpoints, and triangular consecutive triples. Internal degree-five vertices are not excluded by the definition. We prove an explicit two-stage classification. Nonprimitive parameters give no path. For primitive parameters, put $m = \min(l, k-l)$. Apart from unit-width cases, existence is equivalent to $k \equiv \pm 1 \pmod{m}$ when $\lfloor k/m \rfloor \geq 3$, and to $m \equiv \pm 1 \pmod{r}$, where $r = k - 2m$, when that quotient is two. Eight periodic belts give positive witnesses; collisions preceding any admissible endpoint rule out the other cases. Consequences are a smallest primitive negative example $\mathrm{GC}(17,5)[C_{20}] = C_{7980}$ and eventual periodicity along integer affine rays. A density-one statement for negative primitive parameter points remains a separate proof draft. The classification and primitive minimum are Lean-proved on an explicitly specified closed surface graph. The existence of a threshold and positive period is also formalized; effective computation has a prose arithmetic argument. An appendix records a separate finite census of IPR fullerenes from C_{60} to C_{200}: 120,622,224 outputs from each of two generators, with 816 negative isomorphism classes. Novelty relative to the full position-mapping and intersection theory remains unsettled.

1. Introduction

We classify literal linear generalized Stone–Wales paths in the Goldberg–Coxeter images of the dodecahedron. The endpoint condition must hold before the first repeated surface vertex. The proof reduces this comparison to two arithmetic stages, with positive belt witnesses and explicit collisions for the negative cases. The constructions and path definition come from the cited fullerene literature; the exact novelty comparison is discussed in Section 13.

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2. What is proved, and how

Claim	Present proof basis	Scope
Theorem 1	Lean graph theorem and prose geometric proofs	All integer $k > l > 0$, including nonprimitive parameters; explicit closed surface quotient.
Corollary 2	Lean graph and bounded arithmetic theorem	Primitive $k > l > 0$, with the minimum at $(17, 5)$.
Corollary 4	Lean existence theorem; prose threshold arithmetic	Every eventually interior integer affine ray, including nonprimitive points; the executable threshold/period algorithm remains an obligation.
Corollary 3	Prose counting proof draft	Primitive parameter points, with its stated counting and limit obligations.
Finite checks and census	Explicit constructors, path searches and absence-certificate checkers	The ranges in Section 12 and Appendix A; separate full-code comparison through C_{186}.
Auxiliary formal claims	Lean arithmetic and conditional graph propositions	Exact hypotheses and types in the claim ledger.

Two separately developed prose derivations cover the full interior A3 classification. The formal claims concern the specified quotient graph; its interpretation as the intended published construction and human proof peer review remain distinct matters. The priority comparison is open. The supplementary `VERIFICATION-RECORD.md` records roles, independence qualifications, formal entry points and validation history.

3. Objects and definitions

3.1. **The literal path definition.** A fullerene is a simple, three-connected, cubic plane graph all of whose faces are pentagons or hexagons. Its dual is a spherical triangulation with twelve degree-five vertices and all other vertices of degree six. Following Definition 1(a) of [1], a literal gSW path is a sequence $p_0, \ldots, p_e$ satisfying:

(1) $e \geq 3$ is odd and all vertices are distinct;
(2) p_i is adjacent to p_{i+1} and to p_{i+2} whenever defined;
(3) p_0, p_e have degree five, and p_1, p_{e-1} have degree six.

The source's notation $v_1, \ldots, v_{2w}$, $w \geq 2$, gives $e = 2w - 1$. Consecutive adjacency is part of the word "path"; distance-two adjacency is explicit. In particular, internal degree-five vertices are allowed. Requiring an even number of vertices is a definition, not a new necessary condition. "Positive" and "negative" mean presence and absence of this path. Reaction barriers, radial gSW operations, and operations permitting heptagons are outside our scope.

3.2. **The A3 family and lattice coordinates.** Write $A3(k, l) = \mathrm{GC}(k, l)[C_{20}]$, $k > l > 0$. Graver's older coordinates $A3(p, q)$ correspond to $(k, l) = (p + q, p)$. Develop each parent face of the icosahedral dual in the triangular lattice with

$$O = (0, 0), \quad e = (k, l), \quad f = (-l, k + l), \quad T = k^2 + kl + l^2.$$

The fullerene has $20T$ vertices and its dual has $10T + 2$. Let $R(x, y) = (-y, x + y)$ and $G(u, v) = ue + vf$. The maps G and R commute, and the inverse coordinate change is

$$u = \frac{(k + l)x + ly}{T}, \qquad v = \frac{-lx + ky}{T}. \tag{1}$$

In parent coordinates write

$$U(i, j) = [(i, j), (i + 1, j), (i, j + 1)], \quad D(i, j) = [(i + 1, j), (i + 1, j + 1), (i, j + 1)].$$

Developments use integer powers of R and integer translations. A fine unit triangle need not lie in one parent face: pieces cut by parent edges are mapped separately. If $\gcd(k,l)=1$, the vectors $e, f, f-e$ are primitive, so no fine lattice vertex lies inside a parent edge. Identifications of ordinary vertices can therefore be tested in face interiors; parent vertices require separate treatment. Distinct face labels do not imply disjoint closures.

4. The classification

Theorem 1 (Two-stage classification)**.** *Let $k > l > 0$ and $d = \gcd(k,l)$. If $d > 1$, then $A3(k,l)$ is negative. If $d = 1$, put $m = \min(l, k-l)$. The graph is positive if and only if one of the following holds:*

$$\begin{aligned} &m = 1; \\ &m \geq 2, \quad q = \lfloor k/m \rfloor \geq 3, \quad k \equiv \pm 1 \pmod{m}; \\ &m \geq 2, \quad q = 2, \quad r = k - 2m, \quad [\, r = 1 \text{ or } m \equiv \pm 1 \pmod{r}\,]. \end{aligned} \tag{2}$$

Machine-readable claim: `thm:classification`.

For primitive parameters with $m \geq 2$, equality $k = 2m$ is impossible; thus $0 < r < m$ in the last line. No congruence modulo zero is used. We handle $r = 1$ before the remaining congruences.

The proof separates the integer-inflation obstruction (Section 5), the two start orbits (Section 6), positive paths (Section 7), and collisions with endpoint exclusion (Sections 8–9). The four regions in the original coordinates are as follows.

Region	Positive construction	Negative construction
$k > 3l$	AR vertical/horizontal belts	Outer fan
$2l < k < 3l$	AT second belt	Positive fan, $c = 2$
$l < k < 2l$, $k \geq 3(k-l)$	Diagonal belt	Positive fan, $c = 1$
$l < k < 2l$, $k < 3(k-l)$	Folded second belt	Negative fan

Primitive points with $k = 2l$ or $3l$ have $l = 1$ and belong to the unit-width case below. In the third region, $k \bmod (k-l) = l \bmod (k-l)$. In the fourth, $r = 2l - k$ and $m = k - l = l - r$, so $m \bmod r = l \bmod r$. This gives (2), with the assembly in Section 9. The equality of decisions at (k,l) and $(k, k-l)$ follows from the classification, not from a graph isomorphism: their norms differ by $2k(k-2l)$ and their orders generally differ.

5. The obstruction for nonprimitive parameters

Machine-readable claim: `sec:inflation`.

If $d = \gcd(k,l) > 1$, the construction is the integer d-fold inflation of $\mathrm{GC}(k/d, l/d)[C_{20}]$. More generally, in $\mathrm{GC}(d,0)[F]$ the degree-five parent points lie in the d-multiple lattice in each chart. Develop the path from its initial triangle. At an odd position, the determinant of the lattice point with the direction of progression is ± 1. If this point were degree five, that determinant would have to be divisible by d, a contradiction. The gluings are lattice rotations through multiples of 60 degrees and transfers about points of the d-multiple lattice, preserving divisibility. This does not assume a global prohibition on internal degree-five vertices.

The closed quotient retains lattice points on parent edges, and a proved graph isomorphism transports absence to the specified graph. The degree argument uses actual degrees and permits internal degree-five vertices.

6. Two starts and candidate endpoints

Machine-readable claim: `sec:starts`.

6.1. **Forced continuation and the first repeat.** Suppose every edge has exactly two common neighbours. From an oriented triangle (a, b, c), the next vertex is the common neighbour of b, c other than a. Every simple path is a prefix of this forced walk. If ρ is its first repetition index, existence is equivalent to an admissible endpoint before ρ, and $\rho \le |V|$. This conditional principle is a conditional finite-graph theorem. Its local hypothesis and all-start coverage are supplied by the concrete surface argument, which derives the degrees, common-neighbour property and start reduction.

6.2. **The two rotational orbits.** The 60 orientation-preserving parent rotations act on the refined lattice. Cyclic permutations of a face's corners act locally by $R^{2p}z + (0, e, f)_p$ and respect the edge gluings. They are transitive on the twelve parent vertices, with the fivefold stabilizer transitive on the five initial radial edges. The two choices of third vertex have opposite orientation and are not exchanged. Thus the 120 starts split into two orbits of size 60, represented by

$$A = ((0,0),(1,0),(0,1)), \qquad B = ((0,0),(0,1),(1,0)).$$

Their developments are

$$\begin{aligned} A:&\quad P_{2j} = (0, j), \quad P_{2j+1} = (1, j), \\ B:&\quad P_{2j} = (j, 0), \quad P_{2j+1} = (j, 1). \end{aligned} \tag{3}$$

A negative proof must exclude both representatives.

6.3. **The first parent-lattice candidates.** Assume $\gcd(k, l) = 1$ and $l \ge 2$. Since k, l are coprime to T, there is a unique $0 < h < T$ with $kh \equiv l \pmod T$. The first odd parent-lattice candidates in (1) are

$$A : (1, h),\ E_A = 2h + 1; \qquad B : (t_B, 1),\ t_B = T - h - 1,\ E_B = 2t_B + 1. \tag{4}$$

The first positive even candidate has index $2T$. The integer parent coordinates u_A, v_B satisfy

$$\begin{aligned} Tu_A &= k + l + lh, \qquad Tv_B = k - lt_B, \\ ku_A &\equiv 1 \pmod l, \quad k(-v_B) \equiv -1 \pmod l. \end{aligned} \tag{5}$$

These formulas apply before a cone is encountered. They do not prove that the candidates are reachable without repetition. Positive cases require simplicity of the entire prefix; negative cases require a separate exclusion of earlier parent vertices.

7. Positive cases: eight belts and their caps

Machine-readable claim: `sec:positive`.

7.1. **The boundary $l = 1$.** Take the B sequence through $P_{2k+1} = (k, 1) = e$. Its rectangle $0 \le x \le k$, $0 \le y \le 1$ lies in $U(0,0) \cup D(0,-1)$, whose face labels differ. Each face map is injective and no fine vertex lies inside the common parent edge. Only the endpoints are parent vertices, giving an admissible path.

7.2. **Finite belt tables and infinite developments.** Let F_z be the map from a developed parent triangle to its quotient face. A period v_0 means $F_{z+v_0}(x) = F_z(x - v_0)$ on corresponding cells: the affine map, not just the face label, repeats. Equality one period apart and unique reversible edge gluing extend the table to all integer cells.

Belt	Parent strip	v_0	Faces per period
AR A	$0 \le u \le 1$	$(0,5)$	10
AR B	$-1 \le v \le 0$	$(5,0)$	10
AT A	$0 \le v-2u \le 1$	$(2,4)$	12
AT B	$0 \le u+3v \le 1$	$(6,-2)$	12
Diagonal A	$0 \le v-u \le 1$	$(3,3)$	12
Diagonal B	$0 \le u+2v \le 1$	$(6,-3)$	12
Folded second A	$-1 \le v-2u \le 0$	$(2,4)$	12
Folded second B	$-1 \le u+3v \le 0$	$(6,-2)$	12

AR A alternates $U(0,j), D(0,j)$; AR B alternates $D(j,-1), U(j+1,-1)$. A diagonal cell i uses $U(i,i), D(i,i), U(i,i+1), D(i,i+1)$. AT A uses $U(i,j), D(i,j)$ for $j = 2i, 2i+1, 2i+2$, and folded second A uses $j = 2i-1, 2i, 2i+1$. For each B belt rotate the triangle list by R^{-1} and reglue from its specified initial face; do not simply reuse rotated quotient labels.

The faces within a period are distinct, and the parent-vertex label sets on the two boundaries are disjoint. Additional cap faces are distinct from the central faces and from one another. There are 20 phase entries: $5+5$ for AR B, $1+2$ for AT A, $1+3$ for diagonal A, and $1+2$ for folded second B. Their vertices, rotations, and translations are recorded in the cap-separation table and the fixed JSON tables. A finite check covers all eight belts and twenty cap entries. General containment is supplied by the following argument.

7.3. **Endpoints.** Use (3) up to index E in the following table. Unit-width overlaps are handled by a width-one or AR construction. If both signs coincide at width two, choose the plus sign; the minus cap arguments use width at least three.

Condition	Start	Q; endpoint index	Parent endpoint W
$l \ge 2$, $k = ql+1$	A	$q(k+l)+l$; $2Q+1$	$(1,q)$
$l \ge 2$, $k = ql-1$	B	$q(k+l)+l$; $2Q-1$	$(q+1,-1)$
$2l < k < 3l$, $r = k-2l$, $l = sr+1$	B	$s(3k+l)+k$; $2Q+1$	$(3s+1,-s)$
Same region, $l = sr-1$	A	$s(2k+3l)+k+l$; $2Q+1$	$(s,2s+1)$
$l < k < 2l$, $w = k-l$, $l = sw+1$	B	$s(2k+l)+k$; $2Q+1$	$(2s+1,-s)$
Same region, $l = sw-1$	A	$s(k+2l)+k+l$; $2Q+1$	$(s,s+1)$
$l < k < 2l$, $w = 2l-k$, $l = sw+1$	A	$s(2k+3l)-k-l$; $2Q+1$	$(s,2s-1)$
Same region, $l = sw-1$	B	$s(3k+l)-k$; $2Q+1$	$(3s-1,-s)$

For AR, $kQ = qT + \varepsilon l$ and $lQ = T - \varepsilon(k+l)$. For the remaining six rows, respectively,

$$\begin{aligned} rQ+(3k+l) &= T, \quad rQ-(2k+3l) = T, \\ wQ+(2k+l) &= T, \quad wQ-(k+2l) = T, \\ wQ+(2k+3l) &= T, \quad wQ-(3k+l) = T. \end{aligned} \tag{6}$$

In each A row $G(W) = (1,Q)$, and in each B row $G(W) = (Q,1)$, except AR B, where it is $(Q-1,1)$. These integer identities are among the Lean lemmas in Section 11; bounds, simplicity, and legality are not consequences of the identities alone.

7.4. **Containment, including pieces cut by parent edges.** Use $[0,1]\times[0,Q]$ for A and $[0,Q]\times[0,1]$ for B, with the upper horizontal bound $Q-1$ for AR B. These rectangles contain all fine unit triangles used by the paths. Without caps, (1) and (6) put the normal coordinate in the strip interval; for example, second belt B has normal $[rx+(3k+l)y]/T \in [0,1]$.

For AR B, odd points above the strip have $t = 0,\dots,q-1$ and even points below it have $t = Q-q,\dots,Q-1$. The upper triangles have vertices with $t \le q$. The bounds

$$T-k(q+1)-l = k(k-q-1)+kl+l^2-l > 0, \quad k-q-1 = q(l-1)-2 \ge 0$$

put them in the initial face together with the belt. Point reflection $P_{E-j} = P_E - P_j$ covers the terminal cap. For second A, put $n = v-2u$. If $n < 0$, then $x = ru-ln$ and $y \ge 0$ give $0 \le u \le 1/r$,

$-u \le v < 2u$, and $u+v < 3/r \le 1$. This lies in $U(0,0) \cup D(0,-1)$. Reflection $z \mapsto W - z$ sends the $n > 1$ cap back to this initial piece and gives terminal faces $D(s-1,2s), U(s-1,2s+1)$. For diagonal A, use $n = v-u$, $x = wu - ln$ to obtain $0 \le u \le 1/w$, $-u \le v < u$, $u+v < 2/w \le 1$, and terminal faces $D(s-1,s), U(s-1,s+1)$.

For folded second B, $n = u+3v$ and $y = ln - wv$. On the initial $n > 0$ piece,

$$u = \frac{3y - (3l-w)n}{w} < \frac{3}{w} \le 1, \qquad u+v = \frac{2y-kn}{w} < \frac{2}{w} < 1.$$

Both are nonnegative, so the piece lies in $U(0,0) \cup D(0,-1)$; D is central and U additional. Reflection sends n to $-1-n$. These are planar containment arguments, not assertions of a reflection symmetry of the quotient surface. Appendix C gives the terminal pieces explicitly.

7.5. **Simplicity and degree conditions.** Two ordinary vertices identified in the central belt must have the same face phase, so their developed difference is $aG(v_0)$, $a \in \mathbb{Z}$. Across the width-one path rectangle the relevant transverse period magnitudes are $5l, 2r, 3w, 2w$ for AR, second, diagonal, and folded second belts. Each exceeds one, excluding $a \neq 0$; $a = 0$ identifies only the same index of (3). Cap labels are distinct and each face map is injective, excluding further repeats.

Every construction has $Q < T$, excluding positive even parent points. Substituting the possible integer normals $0, 1$ (or $-1, 0$) in (1) and (6) leaves just the odd endpoint. For diagonal B with $w = 1$, the possible normal-zero point $(-2, 1)$ is outside the rectangle. For folded second A with $w = 1$ it would have $u = -1$, again outside. Minus-sign caps use width at least three and contain no other parent point. The endpoints belong to disjoint boundary-label sets. Thus the paths are simple, have even order, and meet all degree conditions.

8. Negative cases: collisions after leaving a belt

Machine-readable claim: `sec:negative`.

The specified collision need not be the first repeat: an earlier repeat is also an obstruction. We must follow every fine triangle up to that time in the stated development and exclude an earlier legal endpoint. Section 9 supplies the latter exclusion.

8.1. **The outer region.** Let $k = ql + r$, $q \ge 3$, $2 \le r \le l-2$, and put

$$a = (q-4)k + (q+2)l - 1, \qquad b = (q+2)k + 6l - 1.$$

A leaves $D(0,q)$ for $U(1,q), D(1,q), U(1,q+1), D(1,q+1)$; the final two face maps compose the old belt map with $H_A(z) = Rz + (q+2, -1)$. B leaves $U(q+1,-1)$ for $D(q+1,-2), U(q+2,-2), D(q+2,-2), U(q+3,-2)$, using $H_B(z) = Rz + (q+1, -q-2)$ on the final two. The rotation centres are $(1, q+1)$ and $(q+2, -1)$. The finite table in the outer collision atlas covers q mod 5. The relations

$$H_A(q+5, z+(0,5)) = H_A(q,z) + (0,5), \quad H_B(q+5, z+(5,0)) = H_B(q,z) + (5,0)$$

and belt periodicity extend it to all q.

The A rectangle $0 \le x \le 1$, $0 \le y \le b$ has $0 \le u < 2$, $v < q+2$. It crosses $u = 1$ at $v = q + (r-x)/l \in (q, q+1)$. For $u \le 1$ one has $v < q+1$; for $u \ge 1$, $v > q$. It is therefore covered by the belt and the four added faces. The last point $(0, b)$ has $1 < u < 2$, $q+1 < v < q+2$. For B, $-2 < v < 1$ and $u + v < q + 2$; its initial $v > 0$ piece lies in $U(0,0)$. At $v = -1$, $u = q + 1 + (r+y)/l \in (q+1, q+2)$, leading into the four lower faces. The last point $(b, 1)$ has $-2 < v < -1$ and $q + 1 < u + v < q + 2$. Neither rectangle crosses the cut beyond the rotation

centre. The following six positive quantities justify these claims:

$$\begin{aligned} 2T-k-l-lb &= (q^2-4)l^2+3qrl+2r^2-ql-r,\\ (q+2)T-kb &= (q-4)kl+(q+2)l^2+k,\\ (q+2)T-kb-k-l &= (q-4)kl+(q+2)l^2-l,\\ lb-T &= k(l-r)+5l^2-l,\\ kb-(q+1)T &= (4q-1)l^2+(q+5)rl+r^2-ql-r,\\ lb-T-k &= k(l-r-1)+5l^2-l. \end{aligned}$$

For $q=3$, substitute $k=3l+r$, $r\le l-2$ in the second and third lines; for $q\ge 4$, positivity follows directly. The collision identities are

$$\begin{aligned} (1,a) &= R(0,b)+(q+2)e-6f,\\ (a,0) &= R(b,1)+(q-4)e-(q+2)f. \end{aligned} \tag{7}$$

These differ from the maps H_A, H_B by the periods $-5f, -5e$. The early A point has $0<u<1$ and the early B point $-1<v\le 0$; for example $T-k-l-la = kb-(q+1)T>0$. Since $0\le a<b$, the actual revisits are $2b\to 2a+1$ for A and $2b+1\to 2a$ for B.

8.2. **The two positive-side fan regions.** Put $k=cl+w$, $l=sw+d$, $2\le d\le w-2$, with $c=1,2$ and $s\ge 2,1$, respectively. The strip $0\le v-cu\le 1$ has period $p(1,c)$, where $p=3,2$, respectively. Set

$$a=(c+1)sk+(s-5)l-1, \qquad b=(cs+5)k+(c+1)sl-1,$$
$$W=(s+1,cs+c+1), \qquad H(z)=R^{-1}z+(-W_v,W_u+W_v).$$

From $D(s,W_v-1)$ pass through $U(s,W_v), D(s,W_v), U(s+1,W_v)$. On the last two faces, the old belt map composed with H is the new map. B uses the rotated triangle list and translation, glued anew. The table in the positive-fan atlas covers every s mod p.

Rotate B's fine and parent coordinates by R only for comparison. In common fine coordinates (X,Y), A has $0\le X\le 1$, $0\le Y\le b$; B has $-1\le X\le 0$, $0\le X+Y\le b$. Since $X=wu-l(v-cu)$, the normal-one crossing is $u=s+(d+X)/w\in(s,s+1)$, before the cut from W. The bounds $v<W_v+1$, $u+v<s+W_v+2$ cover the portion beyond the strip by the original cell and the three added triangles. If $u\le s+1, v\le W_v$, use the original cell; if $v>W_v$, use the new faces; if $u>s+1$, use the final U. The initial negative-normal A piece has $u\le 1/w$, $-u\le v<cu$, $u+v<(c+1)/w\le 3/4$, and lies in $U(0,0)\cup D(0,-1)$. B has no such piece. The last points, A's $(1,b)$ and B's original $(b,0)$, have $v>W_v$, $u+v>s+W_v+1$.

The nine quantities needed for these bounds, the early point, and time ordering are strictly positive:

$$\begin{gathered} a, \quad b-a, \quad T-b,\\ (cs+c+2)T-(kb+k+l),\\ ((c+1)s+c+3)T-((k+l)b+k),\\ kb-l-(cs+c+1)T,\\ (k+l)b-((c+1)s+c+2)T,\\ T-[wa+(c+1)k+l], \qquad (s+2)T-(k+l+lb). \end{gathered} \tag{8}$$

Substitute $s=S+s_{\min}$, $w=D+E+4$, $d=D+2$, with $S,D,E\ge 0$, where $s_{\min}=2,1$. All 18 polynomials have nonnegative coefficients and positive constant terms. Every permitted parameter is obtained by $S=s-s_{\min}$, $D=d-2$, $E=w-d-2$. This is a universal coefficient proof,

not a finite extrapolation. Lean checks the 18 inequalities, while their geometric interpretation is separate. The collision identities are

$$\begin{aligned}(0,a) &= R^{-1}(1,b) - (cs+5)e + (c+1)sf,\\ (a,1) &= R^{-1}(b,0) + (s-5)e + (cs+5)f.\end{aligned} \tag{9}$$

The images differ from H by one belt period. The eighth quantity puts the early point inside the old belt. Thus A revisits $2b+1 \to 2a$, B revisits $2b \to 2a+1$, and $0 < a < b < T$.

8.3. **The negative-side second region.** Assume $3l/2 < k < 2l$, $w = 2l-k$, $l = sw+d$, $s \geq 2$, $2 \leq d \leq w-2$. The strip is $-1 \leq v-2u \leq 0$, with period $(2,4)$. Set

$$a = (2s-5)k + 3sl - 1, \qquad b = 3sk + (s+5)l - 1,$$
$$W = (s+1, 2s+1), \qquad H(z) = Rz + (3s+2, -s-1).$$

Leave $D(s,2s)$ through $U(s+1,2s), D(s+1,2s), U(s+1,2s+1)$. The last two maps use H. B uses the R^{-1}-rotated table. The negative-fan atlas covers s mod 2. Now $X = -wu - l(v-2u)$, so at normal -1, $u = s + (d-X)/w \in (s, s+1)$. The eight positive quantities are

$$\begin{gathered}a, \quad b-a, \quad T-b, \quad (s+2)T - (k+l+lb),\\ (3s+3)T - ((k+l)b+k), \quad lb-k-(s+1)T,\\ (k+l)b - (3s+2)T, \quad T - (wa+2k+3l).\end{gathered} \tag{10}$$

The same nonnegative substitution, with $s = S+2$, proves positivity and is formalized by eight Lean lemmas. It gives $u < s+2$, $u+v < 3s+3$ throughout the rectangle and $u > s+1$, $u+v > 3s+2$ at the last point. Outside the belt, $u \leq s+1$ implies $v > 2s-1$ and $u > s+1$ implies $v > 2s$, using $k(s+1) - 1 - 2sl = (s-1)w + 2d - 1 > 0$ in the latter case. These bounds put every piece in the stated triangles without crossing the cut. A has no initial positive-normal cap; B's initial cap lies in the two faces of Section 7. Finally,

$$\begin{aligned}(1,a) &= R(0,b) + 3se - (s+5)f,\\ (a,0) &= R(b,1) + (2s-5)e - 3sf.\end{aligned} \tag{11}$$

The difference from H is one belt period, giving A's $2b \to 2a+1$ and B's $2b+1 \to 2a$ as actual ordinary-vertex repeats.

9. Endpoint ordering and assembly

Machine-readable claim: `sec:ordering`.

In the outer region, the nonunit residue and (5), with $u_A \geq 1$, $v_B < 1$, imply $u_A \geq 2$, $v_B \leq -2$. Hence

$$lh = Tu_A - k - l \geq 2T - k - l > lb, \qquad lt_B = k - Tv_B \geq k + 2T > lb.$$

Both odd candidates follow the collision. Also $T - b = k(k-q-2) + l(k+l-6) + 1 > 0$, excluding an earlier even parent point.

In the other regions, $k \equiv \pm w \pmod{l}$, $l = sw+d$. For $1 \leq j \leq s$, $w \leq wj \leq l-d \leq l-2$; for $j = s+1$, $wj = l + (w-d)$ with $2 \leq w-d \leq w-2 < l-1$. Thus none of $1, \ldots, s+1$ gives $wj \equiv \pm 1 \pmod{l}$. Equation (5) yields $u_A, -v_B \geq s+2$. The last bound of (8), or the fourth of (10), again gives $h, t_B > b$, while $T - b > 0$ excludes even parent points. Apply this jointly with continuation of the development until a cone: there is no other fine lattice point inside a unit triangle, and no cone inside a parent edge. No hidden cone crossing is assumed away. Both representative starts are excluded in every negative branch; the two-orbit reduction excludes all 120 starts. Combining the positive constructions, the region table, and the nonprimitive obstruction proves the stated classification with the proof basis summarized in the guarantee table.

10. Three consequences

10.1. **The smallest primitive negative example.**

Corollary 2. *Among primitive parameters $k > l > 0$, the smallest negative A3 example is $(17, 5)$, with $T = 399$ and $20T = 7980$.*

Machine-readable claim: `cor:minimum`.

This is not minimality among all fullerenes or among nonprimitive A3 images. All 70 primitive pairs with $T < 399$ satisfy (2); $(17, 5)$ has quotient three and remainder two. Of the 123 pairs with $T \le 673$, the seven negatives are

$$(17, 5), (18, 5), (19, 7), (22, 5), (17, 12), (23, 5), (21, 8).$$

The bound reduces enumeration to $k \le 25$, $0 < l < k$. Of all 123 primitive pairs in this range, 116 are positive. At C_{7980} the two orbit repeats are $228 \to 15$ and $229 \to 14$. The minimum also has the formal proof summarized in Section 11.

10.2. **Density of primitive parameter points.**

Corollary 3. *Let $D(X)$ count primitive pairs $k > l > 0$ with $T \le X$, and let $P(X), Z(X)$ count the positive and negative pairs. Then*

$$\begin{aligned} D(X) &= \frac{X}{\pi\sqrt{3}} + O(\sqrt{X}\log X), \\ P(X) &= \Theta(\sqrt{X}\log X), \\ \frac{Z(X)}{D(X)} &= 1 - \Theta(\log X/\sqrt{X}) \longrightarrow 1. \end{aligned} \tag{12}$$

Machine-readable claim: `cor:density`.

These are parameter counts, not a uniform measure on all fullerenes or on graph isomorphism classes. The following is the prose counting argument; its current proof-draft scope is stated in the guarantee table. Put $K = \lfloor\sqrt{X}\rfloor$. Separate $m = 1$, $k = qm \pm 1$, and the second-stage cases $r = 1$ or $m = sr \pm 1$, $k = (2s + 1)r \pm 2$. There are at most two choices of l for each (k, m), giving, with $H_K = \sum_{j=1}^{K} 1/j$,

$$P(X) \le 4K + (6K + 8)H_K = O(\sqrt{X}\log X).$$

For a lower bound, let $L = \lfloor\sqrt{X/3}\rfloor$ and $M = \lfloor(L - 1)/3\rfloor$. Count $(k, l) = (qm + 1, m)$ with $2 \le m \le M$ and $3 \le q \le \lfloor(L - 1)/m\rfloor$. These points are primitive and positive and satisfy $k \le L$, $T \le 3L^2 \le X$. Their number is $(L - 1)(H_M - 1) + O(M) = \Omega(\sqrt{X}\log X)$. The region $k > l > 0$, $T \le X$ has area $\pi X/(6\sqrt{3})$, with lattice-point error $O(\sqrt{X} + 1)$. Möbius inversion, $\sum \mu(d)/d^2 = 6/\pi^2$, gives the formula for $D(X)$ with error $O(\sqrt{X}\log X)$, and hence (12).

10.3. **Eventual periodicity on integer affine rays.**

Corollary 4. *Let $k(t) = at + b$, $l(t) = ct + d$ have integer coefficients and satisfy $k(t) > l(t) > 0$ for all sufficiently large integer t. The existence sequence is eventually periodic, with an arithmetically computable threshold and period.*

Machine-readable claim: `cor:rays`.

This is a consequence of Theorem 1, not a separate novelty claim. If $\Delta = ad - bc \ne 0$, then $\gcd(k, l) = \gcd(k, l, \Delta)$, so primitivity depends only on $t \bmod |\Delta|$. If $\Delta = 0$, a nonconstant ray is an integer multiple of a fixed primitive direction, with multiplier eventually at least two, and is negative. A constant ray is constant. Eventually $m = \min(l, k - l)$ is one affine expression. Constant m gives fixed-modulus congruences. If m grows, $\lfloor k/m\rfloor$ stabilizes; even when the limiting ratio is an integer, the constant sign of the remainder selects the eventual quotient. At quotient

two, $r = k - 2m$ is affine; persistent $r = 0$ is nonprimitive and $r = 1$ is positive. The remaining conditions are affine divisibility tests.

For $A(t) = \alpha t + \beta > 0$, $B(t) = \gamma t + \delta$ with $\alpha > 0$, $A \mid B$ implies $A \mid (\alpha\delta - \gamma\beta)$. If that constant is nonzero, divisibility eventually fails. If it is zero, $B = (\gamma/\alpha)A$ and divisibility holds identically exactly when γ/α is an integer. For $\alpha = 0$ it is a fixed-modulus congruence. Combine the inequality thresholds and take the least common multiple of the fixed moduli and $|\Delta|$. After classification, the conclusion also falls under parametric Presburger arithmetic [9].

11. Formal graph and proved conclusions

The graph specification uses twenty oriented parent faces on twelve labels, integer lattice points of closed triangles, and equivalence generated by edge gluings. The simple graph `gcGraph` defines adjacency from lattice triangles. Its `LiteralPath` uses actual neighbour sets, distinct vertices and an even vertex count, permitting internal degree-five vertices. The predicate `criterion` includes primitivity and the alternatives in (2), on natural inputs $0 < l < k$.

Proved graph isomorphisms connect the surface descriptions to this quotient, retaining ordinary parent-edge vertices in the nonprimitive case. Degree five is equivalent to being a parent vertex; other vertices have degree six. The argument derives local triangle continuation, coherent chart changes and a finite lift of every literal path. Primitive necessity uses start reduction and sector collisions; positive constructions give the converse. The full theorem is

$$(\exists n, p,\ \mathrm{LiteralPath}(\mathrm{gcGraph}(k, l), n, p)) \quad \Longleftrightarrow \quad \mathrm{criterion}(k, l).$$

There is no primitivity hypothesis. For Corollary 2, the formal bounded arithmetic theorem shows that among primitive $k > l > 0$ with $T \leq 399$, absence is equivalent to $(k, l) = (17, 5)$. Its supporting lemmas prove absence, existence below the bound, primitivity and the norm calculation.

For Corollary 4, the formal arithmetic result supplies a threshold and positive period on every eventually interior integer affine ray. The full classification transports it to actual path existence, including nonprimitive points. A natural ray parameter describes the same positive tail as sufficiently large integer time. The prose argument above gives the effective arithmetic construction; the formal result has existence scope. Auxiliary claim identifiers below refer to their exact typed statements in the claim ledger.

12. Finite checks

12.1. **Parameter checks and path audits.** An explicit graph constructor and literal-path checker test the classification on 123 primitive pairs through $T = 673$: 116 positive and seven negative. A discovery set through $T = 5000$ has 921 points, 602 positive and 319 negative. Further fixed predictions include sets of twenty, fourteen, forty-six and thirty points; all decisions match. The forty-six-point range is $9049 \leq T \leq 185851$ and the thirty-point maximum is $T = 488721$. A separately constructed comparison of 360 pairs with $5000 < T \leq 7000$ has 171 positive and 189 negative decisions; twenty-six were known, leaving 334 new inputs.

Two arithmetic implementations agree at cutoffs $10^4, 10^5, 10^6, 10^7$. At $X = 10^7$ there are 1,837,762 primitive points, 81,946 positive and 1,755,816 negative (about 95.54%). These are finite arithmetic counts. Direct endpoint formulas agree with unit continuation on 161 unit-residue, 76 second-stage and 323 assembled paths, totalling 560 paths with repeated graphs.

The finite belt audit checks eight belts, twenty caps and 337 endpoint orders (159 outer, 63 second positive fan, 68 first positive fan, 47 folded second). It checks every vertex, cut triangle piece and cut on 160 known negative graphs and 320 starts, and all 19,200 starts of those same graphs. Nine rays have 1,809 arithmetic comparisons and fourteen graph instances. The aggregate workload is 286,940 vertices, 286,300 triangles, 303,857 pieces and 19,412 cuts; these counts overlap the examples rather than enlarging their number.

A separate constructor checks 191 theorem-domain pairs with $T \le 700$: 127 primitive (120 positive, seven negative) and 64 nonprimitive, all negative. Another 39 nonprimitive boundary pairs consist of 25 $(k,0)$ and 14 (k,k) inputs. Two controls $(1,0),(1,1)$ give 232 stored records in total. All decisions agree, including minimum $T = 399$; positive paths and 13,200 negative-start certificates are checked. The boundary pairs are outside Theorem 1.

The IPR census has a separate full-canonical-code-set comparison on 64 even sizes C_{60}–C_{186}, with 43,636,843 outputs per generator and 346 negatives. The full C_{60}–C_{200} totals and the scope of their decision and certificate checks are given in Appendix A.

12.2. **A second geometric derivation.** The negative proof reduces the 120 starts to two orbits of 60 under icosahedral rotations. Its developed rows are, for integer t,

$$p^+_{2t} = (t,0), \quad p^+_{2t+1} = (t,1), \qquad p^-_{2t} = (t,0), \quad p^-_{2t+1} = (t+1,-1).$$

After the belt region, the other sectors use four- or six-face blocks. Face-label permutations of order three or two identify a twelve-face period. The proof supplies face containment, gluing and exclusion of intermediate parent vertices. With the sector-dependent integers A, J defined by the corresponding sector developments, the return maps are written here as $\mathcal{G}, \mathcal{H}$ to distinguish them from our GC coordinate map:

$$\mathcal{G}(x,y) = (A - y, x + y - J), \qquad \mathcal{H}(x,y) = (x + y + A - J, -x + J).$$

They give the following explicit equalities of surface vertices:

	upper row	lower row
$\mathcal{G}$	$2J-1 \mapsto 2A-2$	$2J-2 \mapsto 2A-1$
$\mathcal{H}$	$2J-2 \mapsto 2A-1$	$2J-1 \mapsto 2A-2$

The sector inequalities put a repeated vertex before any legal endpoint; the argument needs such a repeat, not a proof that it is the first repeat. Together with the belt and nonprimitive arguments this excludes all negative parameters, including paths that might otherwise meet an internal degree-five vertex.

For the remaining positive sectors, the construction has endpoints $E = (A, \sigma)$, $\sigma = \pm 1$, in a strip of width one. Four-face blocks with an order-three permutation, or six-face blocks with an order-two permutation, run through twelve distinct parent faces. Copies of a face are separated transversely by $-3d, -2d$, or $2d$ with $d \ge 2$. The proof checks the initial and terminal caps as well; primitive parent edges have no interior lattice points. These facts give injectivity, distinct endpoint labels, actual endpoint degrees, and both consecutive and skip adjacency. The two row constructions have $2A+2$ or $2A$ vertices, respectively. Small widths and boundary ratios are handled by the previously proved belt branch.

The face tables and repeat formulas give the second geometric route; both routes use developed strips, belts and transverse width.

12.3. **Additional checks and the $T = 1099$ separation.** An additional finite audit verifies 24 gluing records (ten outer, ten positive-side and four negative-side), 82 edge moves, 48 terminal triangles and 144 vertex images. It checks 560 known positive paths on 417 distinct graphs, comprising 584,420 vertices and 583,300 unit triangles, and 159 known outer-negative graphs with 318 starts. The latter checks include repeats, prior-cone exclusion, regions, cuts and edge gluings. Probes overlap these samples.

The pairs $(23,15)$ and $(27,10)$ have $T = 1099$, the same final group product u on 60 states, and six closed zigzags of length 10990. Formula (2) nevertheless labels the first positive and the second negative. Explicit constructions of both C_{21980} graphs (10,992 dual vertices, IPR) give a 290-vertex path in the first and no path in the second. The pair is a known comparison example. It shows that existence is not a function of (T,u) alone. It does not exclude a derivation using the full position mapping, z-vectors, or intersection order. The comparison finds that the type-I/type-II

self-intersection signatures differ: $(320, 590)$ versus $(270, 640)$. Thus this example separates only the compared counts, lengths and pentagon-contact data, not the full intersection theory.

13. Related work and open comparison

13.1. **Related work and the novelty boundary.** The problem comes from the classification question for fullerenes admitting no generalized Stone–Wales operation in [1]. Our literal Definition 1(a) scope and the restriction to interior A3 parameters are essential. The Goldberg–Coxeter construction, tightness of primitive images, and arithmetic of closed zigzags are established theory. Dutour–Deza's position mapping uses a directed parent-dual edge and a position p; its group product determines zigzag number and length and can be computed by a Euclidean algorithm [2, Section 4, Theorems 4.5, 4.12, 6.5 and Proposition 4.14]. For primitive dodecahedral images, the normalized partition vector has three possible types. Zigzags, railroads, and their self- and mutual intersections have also been studied [3, 8]. None of these constructions or methods is itself claimed as new here.

The compared counts, lengths and pentagon-contact sequences alone do not determine the existence of a simple open gSW path. In a comparison of 489 already known primitive pairs, adding the parent-distance sequence of pentagon contacts or the counts of shared pentagons still did not separate the two-stage decisions. At $T = 1099$, even the complete pentagon-contact cycles of all zigzags, in the fixed parent labels and up to cyclic start and reversal, agreed for the pair above. However, the check found different self-intersection signatures with types I and II kept separate: $(320, 590)$ and $(270, 640)$, respectively. This is not a counterexample to determination from the complete z-vector or full intersection order. Two tracing implementations agree on five known examples using a shared graph constructor. The 489 labels come from the classification formula rather than a new graph census.

Full timing information can recover the input. Put $n = k+l$ and let τ be the first position-update count from $p = 1$ to $p = n$. Then $\tau l \equiv -1 \pmod{n}$, so the unordered adjacent intervals $\{\tau, n - \tau\}$ determine (k, l) under $k > l > 0$. Recovering the input alone is not a short deduction of gSW existence. The additional geometric task is to preserve legal endpoints, parity, and the order of revisits to original dual vertices, and reduce their stopping condition to the two residue stages.

Comparing self-intersection against arrival at the next pentagon also has precedent: [4, Section 6, Fig. 5] uses such a comparison for Coxeter coordinates $(7, 2)$, C_{1340}, to exclude a particular twisted Endo–Kroto reduction patch. That path and operation differ from literal gSW. Catalogues of isomerization patches [5], Caspar–Klug T- and gcd-based classifications [6], and congruence conditions in a pentagonal-polyhedron model [7] are related, but the portions examined did not supply this same gSW criterion in the portions compared. The exact reading scope is recorded in the supplement. The comparison with Deza–Dutour–Fowler's zigzag and railroad theory [8] leaves open whether the criterion is a short consequence of the full intersection theory. The candidate contribution is the reduction, over all interior A3 parameters, of the comparison between the first legal pentagonal endpoint and the first revisit to the same original face to an explicit two-stage residue criterion. More precisely, from a legal start let R be the first repeated original face index and E the first odd index at least three with a pentagonal face preceded by a hexagonal face, before imposing simplicity ($E = \infty$ if none exists). A literal path exists from that start exactly when $E < R$. Internal pentagons are allowed. Known flag or position-mapping encodings already permit a finite search; the candidate difference is the closed arithmetic condition for some legal start, with geometric positive witnesses and negative revisits. Neither GC construction, gcd decomposition, Euclidean group products nor finite decidability is proposed as a new principle. Priority and a possible short derivation from full position-mapping theory remain open. After classification, ray periodicity is a consequence of parametric Presburger arithmetic [9, Theorem 1.15, Property 1], not a new general periodicity principle.

Appendix A. IPR census from C_{60} through C_{200}

Machine-readable claim: `app:census`.

A.1. **Scope, method, and preservation.** IPR means that no two pentagons share an edge. The 71 even sizes $N = 60, 62, \ldots, 200$ were enumerated with buckygen 1.1 and fullgen from plantri 5.8. The cumulative record contains 120,622,224 outputs per generator. The pipeline checked the dual, spherical rotation system, degrees, IPR, and the classifier's local prerequisites. Negative canonical-code sets agreed at every size, giving 816 distinct classes. Full canonical-code sets of all positive graphs are compared only on the separate range through C_{186}. Using two generators does not independently validate their shared decision code.

Before adopting the C classifier, the C–Python audit compared it with both forced continuation and a definition-based DFS on 1,456,598 general C_{20}–C_{100} outputs and 4,770,156 IPR C_{60}–C_{160} outputs: 6,226,754 in total, all agreeing, with positive witnesses checked against the definition. These were buckygen outputs and the two ranges overlap. This is not a Python rerun of all 120,622,224 outputs. Subsequent negatives were checked by both Python searches and exhaustive-start absence certificates. Generator completeness and correctness on every possible input are not formally proved by these tests.

Saved records include size/generator/partition counts and completion, implementation hashes, negative full codes and adjacency lists, 16 audits and cumulative summaries, and one JSON certificate per negative class. A certificate contains the dual adjacency and the maximal simple forced path and stopping reason for every admissible initial triangle. The index's `canonical_sha256` hashes canonical-code bytes; `certificate_sha256` hashes the JSON file. The original audits compare the full codes, with metadata extraction checking saved hashes and counts. The census includes discovery and later confirmation; fixed predictions cover the negative sets for C_{186}–C_{200} and match the observed sets.

A.2. **Counts by size.** The following table gives the saved counts from `data/counts.csv`. Zero total output and zero negatives among existing graphs are distinct.

Carbon order	Outputs per generator	Negative classes
60	1	0
62	0	0
64	0	0
66	0	0
68	0	0
70	1	0
72	1	0
74	1	0
76	2	0
78	5	0
80	7	1
82	9	0
84	24	0
86	19	0
88	35	0
90	46	0
92	86	0
94	134	0
96	187	1
98	259	0

Carbon order	Outputs per generator	Negative classes
100	450	0
102	616	0
104	823	1
106	1,233	0
108	1,799	0
110	2,355	0
112	3,342	2
114	4,468	0
116	6,063	0
118	8,148	0
120	10,774	3
122	13,977	0
124	18,769	0
126	23,589	0
128	30,683	6
130	39,393	0
132	49,878	0
134	62,372	0
136	79,362	6
138	98,541	0
140	121,354	0
142	151,201	0
144	186,611	15
146	225,245	0
148	277,930	0
150	335,569	0
152	404,667	17
154	489,646	0
156	586,264	0
158	697,720	0
160	836,497	40
162	989,495	0
164	1,170,157	1
166	1,382,953	0
168	1,628,029	45
170	1,902,265	1
172	2,234,133	0
174	2,601,868	0
176	3,024,383	89
178	3,516,365	0
180	4,071,832	1
182	4,690,880	0
184	5,424,777	117
186	6,229,550	0
188	7,144,091	0
190	8,187,581	0
192	9,364,975	199
194	10,659,863	0

Carbon order	Outputs per generator	Negative classes
196	12,163,298	0
198	13,809,901	0
200	15,655,672	271
Total	120,622,224	816

A.3. **The 816 classes and their relation to A3.** The finite taxonomy assigns disjoint explanations:

Explanation	Classes
GC$(2,0)$ images	812
GC$(3,0)[C_{20}]$, at C_{180}	1
$C1(2,3) = F(2,3)$, at C_{164}	1
$O2(2,3) = G(2,2,3)$, at C_{170}	1
$C1(3,2) = F(3,2)$, at C_{184}	1
Unexplained in this finite census	0

The GC2 comparison includes 116 at C_{184}, 199 at C_{192}, and 271 at C_{200}. The three noninteger-GC examples were identified by full canonical-code equality with their constructions. Here "noninteger GC" means not GC$(d,0)[F]$ for an ordinary fullerene F and $d \geq 2$, not the complement of all GC images. The C1 and O2 identifications are with existing catalogue families, not new family constructions. Their general absence arguments are separate from the individual certificates and this finite accounting.

In the theorem's interior domain, $20T \leq 200$ permits only $(2,1)$, $T = 7$, C_{140}, which is IPR and positive, consistently with $m = 1$. All 121,354 IPR C_{140} outputs are positive. The boundary points $C_{60} = \mathrm{GC}(1,1)[C_{20}]$, $C_{80} = \mathrm{GC}(2,0)[C_{20}]$, and $C_{180} = \mathrm{GC}(3,0)[C_{20}]$ are not interior-theorem cases. The census therefore does not verify a broad range of interior A3 parameters.

A.4. **Independent comparison and the local data package.** The separate full-code comparison covers C_{60}–C_{186}: 64 even sizes, 43,636,843 outputs per generator and 346 negatives. The C_{188}–C_{200} results use the primary census pipeline. The data package contains an English README and schema, a 71-row CSV, an 816-row JSONL index, 816 certificate JSON files, audit summaries, decision and checking code with dependency hashes, and generator metadata. The certificates total 60,758,514 uncompressed bytes. Generator source and binaries, raw output streams and all positive graphs are excluded. Package-relative references retain original and derived hashes. Third-party terms are retained.

Appendix B. The negative B orientation: coordinates, cuts, and ownership

Machine-readable claim: `app:B`.

This appendix expands Section 8 without new parameter conditions. It covers every piece of every fine triangle, not just vertices. Assign piece interiors to parent faces and use the agreement of neighbouring affine gluing maps on boundaries. The facial-triangle prerequisite, primitive parent edges, and the exclusion of earlier parent vertices are as in the body and Section 9.

B.1. **Two coordinate systems.** Let (x, y) be B's original fine coordinates and (α, β) its original parent coordinates. Its triangle sequence lies in $0 \le x \le b$, $0 \le y \le 1$, with $T\alpha = (k+l)x + ly$, $T\beta = -lx + ky$. For the two slanted fan types the actual faces are rotated by R^{-1}. For calculation only, rotate back to common coordinates:

$$\begin{gathered}(X, Y) = (-y, x+y), \qquad (u, v) = (-\beta, \alpha+\beta),\\ Tu = lx - ky, \quad Tv = kx + (k+l)y, \quad T(u+v) = (k+l)x + ly.\end{gathered} \tag{B1}$$

Thus $-1 \le X \le 0$, $0 \le X+Y \le b$, not A's rectangle. The original parent-face names are recovered by

$$R^{-1}U(i,j) = D(i+j, -i-1), \quad R^{-1}D(i,j) = U(i+j+1, -i-1). \tag{B2}$$

This planar coordinate change does not rotate quotient labels; use the actual B gluing tables for each face map.

B.2. **Outer fan: $k = ql + r$.** Here retain (α, β), with strip $-1 \le \beta \le 0$. Call the six positive expressions in Section 8 $F_1, \dots, F_6$. Throughout the rectangle,

$$\beta \ge -lb/T > -2, \quad \beta \le k/T < 1, \quad \alpha + \beta \le (kb + k + l)/T < q + 2.$$

The first bound uses $2T - lb = F_1 + k + l$ and the last uses F_3. If $\beta > 0$, then $x < ky/l$, $\alpha \ge 0$, and $\alpha + \beta = [kx + (k+l)y]/T < y/l \le 1/l < 1$, placing the initial piece in $U(0,0)$. At $\beta = 0$, $\alpha = y/l$. At $\beta = -1$,

$$\alpha = q + 1 + (r+y)/l \in (q+1, q+2),$$

using $r \ge 2$, $r + 1 \le l - 1$; the cut beyond $(q+2, -1)$ is avoided. If $\beta < -1$, then $\alpha + \beta = (y - k\beta)/l > k/l > q$. The resulting region $-2 < \beta < -1$, $q < \alpha + \beta < q + 2$ is covered by

$$D(q+1, -2),\ U(q+2, -2),\ D(q+2, -2),\ U(q+3, -2).$$

The line $\alpha + \beta = q + 1$ splits the first pair from the second; $\alpha = q+2$ or $q+3$ identifies the individual triangle. The first crossing is the common edge of $U(q+1, -1)$ and $D(q+1, -2)$. At $(b, 1)$,

$$T(-1-\beta) = F_6 > 0, \qquad T(\alpha + \beta - q - 1) = F_5 + k + l > 0.$$

Thus H_B applies on the last two faces. Its translation differs from the B translation in (7) by $-(5, 0)$. The early point $(a, 0)$ has $\beta = -la/T$ and $T - la = F_5 + k + l > 0$, so it is inside the old belt. Together with ordering and Section 9, this gives $2b + 1 \to 2a$.

B.3. **Positive-side fans: $k = cl + w$, $l = sw + d$.** Let $c = 1, 2$, with $s \ge 2, 1$, $2 \le d \le w - 2$, and $W_v = cs + c + 1$. The common normal is $n = v - cu = \alpha + (c+1)\beta$, with

$$Tn = wx + [(c+1)k + l]y \ge 0, \qquad -y = wu - ln.$$

There is no initial negative-normal B piece. At $n = 1$, $u = s + (d - y)/w \in (s, s+1)$. For $n > 1$, $u > (l-1)/w > s$, so exit is on the cell-s side of the cut from W. Call the quantities (8) $P_1, \dots, P_9$. Their exact uses are:

Required bound	Positive numerator
$v < W_v + 1$	P_4
$u + v < s + W_v + 2$	$P_5 + (k - l)$
$u < s + 2$	$P_9 + k + l$
$v > W_v$ at $(b, 0)$	$P_6 + l$
$u + v > s + W_v + 1$ at $(b, 0)$	P_7
$n < 1$ at $(a, 1)$	P_8
$0 < a < b < T$	P_1, P_2, P_3

For complete ownership, partition the $n > 1$ portion into three cases. If $u \le s + 1, v \le W_v$, then $s < u \le s + 1$ and $cu + 1 < v \le cs + c + 1$, within the original full cell triangles $U(s, j), D(s, j)$, $j = cs, \dots, cs + c$. Here "cell" includes the full glued triangles, not just their intersection with the slanted strip. If $u \le s + 1, v > W_v$, the bound $v < W_v + 1$ gives $U(s, W_v)$ or $D(s, W_v)$, divided by $u + v = s + W_v + 1$. If $u > s + 1$, then $v > W_v$ and $u + v < s + W_v + 2$, giving $U(s + 1, W_v)$. This covers every piece. The last point belongs to the latter two faces. In B's original coordinates the departure and added faces are

$$U(s + W_v, -s - 1);$$
$$D(s + W_v, -s - 1),\ U(s + W_v + 1, -s - 1),\ D(s + W_v + 1, -s - 2).$$

Conjugating the common map gives $H_B(z) = R^{-1}z + (s+1, W_v)$. The difference from the translation $(s - 5, cs + 5)$ in (9) is $(-6, 4 - c)$: minus one B period, $(6, -3)$ or $(6, -2)$. At $(a, 1)$, $0 < n < 1$, so the early point uses the same belt map. This establishes $2b \to 2a + 1$ without an appeal to analogy with A.

B.4. **Negative-side fan: $k = 2l - w$, $l = sw + d$.** Here $s \ge 2$, $2 \le d \le w - 2$, and

$$n = v - 2u = \alpha + 3\beta, \quad Tn = -wx + (3k + l)y, \quad -y = -wu - ln.$$

At $n = -1$, $u = s + (d + y)/w \in (s, s + 1)$, using $d + 1 \le w - 1$. For $n < -1$, $u > (l - 1)/w > s$, before the cut from $(s + 1, 2s + 1)$. Writing (10) as $M_1, \dots, M_8$, the whole rectangle has $u < s + 2$ from $M_4 + k + l$ and $u + v < 3s + 3$ from $M_5 + (k - l)$. At $(b, 1)$ the lower bounds are M_6 for $u > s + 1$ and $M_7 + l$ for $u + v > 3s + 2$. At $(a, 0)$, $T - wa = M_8 + 2k + 3l > 0$ gives $n > -1$; M_1, M_2, M_3 order the indices.

Since $v = (ku + y)/l$, $u > s$ implies $v > 2s - 1 + d/l > 2s - 1$. For $u > s + 1$,

$$v > k(s + 1)/l = 2s + [(s - 1)w + 2d]/l > 2s.$$

If $n < -1, u \le s+1$, then $2s-1 < v < 2u-1 \le 2s+1$, within $U(s, j), D(s, j)$ for $j = 2s-1, \dots, 2s+1$. If $u > s+1$, the bounds put the point in $U(s+1, 2s), D(s+1, 2s), U(s+1, 2s+1)$: below $v = 2s+1$, split at $u + v = 3s + 2$; above it, use the last U. The last point is in the final two faces. Under (B2), these faces are

$$D(s, 2s) \mapsto U(3s + 1, -s - 1),$$
$$U(s + 1, 2s) \mapsto D(3s + 1, -s - 2),$$
$$D(s + 1, 2s) \mapsto U(3s + 2, -s - 2),$$
$$U(s + 1, 2s + 1) \mapsto D(3s + 2, -s - 2).$$

For the initial $n > 0$ piece, return to original coordinates. From $y = ln - w\beta$,

$$0 \le \alpha = \frac{3y - (3l - w)n}{w} < 3/w \le 3/4, \qquad 0 \le \alpha + \beta = \frac{2y - kn}{w} < 2/w \le 1/2.$$

It lies in $U(0, 0) \cup D(0, -1)$, with D central and U additional. Assign $-1 \le n \le 0$ to the central belt. This covers the rectangle. The original map is $H_B(z) = Rz + (2s + 1, -3s - 2)$. The translation

in (11) differs by $(-6,2) = -R^{-1}(2,4)$. The early point has $-1 < n = -wa/T < 0$, proving $2b+1 \to 2a$ at the same ordinary vertex.

Appendix C. Terminal positive caps by point reflection

Machine-readable claim: `app:C`.

C.1. **Action on the sequence and parent triangles.** For $W = (A, B)$ let $J_W(z) = W - z$. In fine coordinates this is $(1, Q) - (x, y)$ for A, $(Q, 1) - (x, y)$ for B, and $(Q-1, 1) - (x, y)$ for AR B. It preserves the relevant rectangle and gives $P_{E-j} = P_E - P_j$. Consecutive triangles and their cut pieces map to the terminal triangles in reverse order. Direct substitution gives

$$J_W U(i,j) = D(A-i-1, B-j-1), \quad J_W D(i,j) = U(A-i-1, B-j-1). \tag{C1}$$

This is planar geometry, not quotient reflection symmetry. Terminal face maps and labels are taken from the fixed cap table.

C.2. **AR B, $k = ql - 1$.** Here $W = (q+1, -1)$, $n = v$, and $n(J_W z) = -1 - n(z)$. The initial $n > 0$ piece has $u \geq 0$, $v > 0$, $u + v \leq y/l \leq 1/l < 1$, so lies in $U(0,0)$. For the terminal $n < -1$ piece, $z' = J_W z$ satisfies

$$u' = q + 1 - u \geq 0, \quad v' = -1 - v > 0, \quad u' + v' = q - u - v \leq 1/l < 1.$$

Thus the terminal cap is $D(q, -2)$, while the initial central $D(0, -1)$ maps to central $U(q, -1)$. Their shared edge is $[(q, -1), (q+1, -1)]$. Reflection preserves the central normal interval. The initial U phase zero and terminal D phase $q - 1 \bmod 5$ have labels distinct from the central belt and each other. Width one was handled earlier; choose the plus branch at width two. Containing each cut piece does not require all vertices of a triangle touching that piece to lie in the same cap face.

C.3. **Second A and diagonal A, $l = sw - 1$.** For $c = 2$, $w = k - 2l$; for $c = 1$, $w = k - l$. Set $n = v - cu$ and $W = (s, cs + 1)$. The minus branch uses $w \geq 3$, and $n(J_W z) = 1 - n(z)$. On the initial $n < 0$ piece,

$$0 \leq u < 1/w, \qquad -u \leq v < cu, \qquad u + v < (c+1)/w \leq 1.$$

It lies in central $U(0,0)$ and additional $D(0, -1)$. Reflecting a terminal $n > 1$ point gives

$$0 \leq s - u < 1/w, \quad -(s-u) \leq cs + 1 - v < c(s-u),$$
$$(c+1)s + 1 - u - v < (c+1)/w \leq 1.$$

The terminal central and additional faces are $D(s-1, cs)$ and $U(s-1, cs+1)$, sharing $[(s-1, cs+1), (s, cs+1)]$. The extra-face phases are $s - 1 \bmod 2$ for second A and $s \bmod 3$ for diagonal A. The fixed table exhausts them. Rectangle invariance and piecewise reflection cover every terminal triangle piece.

C.4. **Folded second B, $l = sw - 1$, $w = 2l - k \geq 3$.** Here $W = (3s - 1, -s)$, $n = u + 3v$, and $n(J_W z) = -1 - n(z)$. For the initial $n > 0$ piece with $0 \leq y \leq 1$,

$$0 \leq u = \frac{3y - (3l - w)n}{w} < 3/w \leq 1, \quad 0 \leq u + v = \frac{2y - kn}{w} < 2/w < 1.$$

Thus $U(0,0)$ is additional and $D(0, -1)$ central. Reflecting the terminal $n < -1$ piece gives

$$0 \leq 3s - 1 - u < 3/w, \quad 0 \leq 2s - 1 - u - v < 2/w, \quad n(z') > 0.$$

The terminal additional face is $D(3s-2, -s-1)$ and the central face $U(3s-2, -s)$, sharing $[(3s-2, -s), (3s-1, -s)]$. The two phases $s \bmod 2$ occur in the table. The central normal interval is preserved and boundary pieces glue along the shared edge.

C.5. **From containment to legality on the surface.** This appendix proves cap containment and gluing as a prose argument. Distinct face labels still depend on the finite table; exclusion of periodic identifications on Section 7; and degrees on $Q < T$ and the candidate congruences.

Auxiliary formal claim identifiers

The following identifiers point to exact typed statements in the claim ledger: 41 arithmetic propositions, a conditional finite-path theorem and three selected prototype propositions.

`unit_endpoint`
Machine-readable claim: `lean:unit_endpoint`.
`second_plus`
Machine-readable claim: `lean:second_plus`.
`second_minus`
Machine-readable claim: `lean:second_minus`.
`diagonal_plus`
Machine-readable claim: `lean:diagonal_plus`.
`diagonal_minus`
Machine-readable claim: `lean:diagonal_minus`.
`folded_second_plus`
Machine-readable claim: `lean:folded_second_plus`.
`folded_second_minus`
Machine-readable claim: `lean:folded_second_minus`.
`outer_collisions`
Machine-readable claim: `lean:outer_collisions`.
`positive_fan_collisions`
Machine-readable claim: `lean:positive_fan_collisions`.
`negative_fan_collisions`
Machine-readable claim: `lean:negative_fan_collisions`.
`no_small_multiple`
Machine-readable claim: `lean:no_small_multiple`.
`outer_u_bound`
Machine-readable claim: `lean:outer_u_bound`.
`outer_v_bound`
Machine-readable claim: `lean:outer_v_bound`.
`candidate_order`
Machine-readable claim: `lean:candidate_order`.
`outer_endpoint_order`
Machine-readable claim: `lean:outer_endpoint_order`.
`au_c1_a_pos`
Machine-readable claim: `lean:au_c1_a_pos`.
`au_c1_b_minus_a_pos`
Machine-readable claim: `lean:au_c1_b_minus_a_pos`.
`au_c1_N_minus_b_pos`
Machine-readable claim: `lean:au_c1_N_minus_b_pos`.
`au_c1_v_upper_pos`
Machine-readable claim: `lean:au_c1_v_upper_pos`.
`au_c1_sum_upper_pos`
Machine-readable claim: `lean:au_c1_sum_upper_pos`.
`au_c1_v_late_lower_pos`
Machine-readable claim: `lean:au_c1_v_late_lower_pos`.

au_c1_sum_late_lower_pos
Machine-readable claim: lean:au_c1_sum_late_lower_pos.
au_c1_early_inside_pos
Machine-readable claim: lean:au_c1_early_inside_pos.
au_c1_endpoint_bound_pos
Machine-readable claim: lean:au_c1_endpoint_bound_pos.
au_c2_a_pos
Machine-readable claim: lean:au_c2_a_pos.
au_c2_b_minus_a_pos
Machine-readable claim: lean:au_c2_b_minus_a_pos.
au_c2_N_minus_b_pos
Machine-readable claim: lean:au_c2_N_minus_b_pos.
au_c2_v_upper_pos
Machine-readable claim: lean:au_c2_v_upper_pos.
au_c2_sum_upper_pos
Machine-readable claim: lean:au_c2_sum_upper_pos.
au_c2_v_late_lower_pos
Machine-readable claim: lean:au_c2_v_late_lower_pos.
au_c2_sum_late_lower_pos
Machine-readable claim: lean:au_c2_sum_late_lower_pos.
au_c2_early_inside_pos
Machine-readable claim: lean:au_c2_early_inside_pos.
au_c2_endpoint_bound_pos
Machine-readable claim: lean:au_c2_endpoint_bound_pos.
au_neg_a_pos
Machine-readable claim: lean:au_neg_a_pos.
au_neg_b_minus_a_pos
Machine-readable claim: lean:au_neg_b_minus_a_pos.
au_neg_N_minus_b_pos
Machine-readable claim: lean:au_neg_N_minus_b_pos.
au_neg_u_upper_pos
Machine-readable claim: lean:au_neg_u_upper_pos.
au_neg_sum_upper_pos
Machine-readable claim: lean:au_neg_sum_upper_pos.
au_neg_u_late_lower_pos
Machine-readable claim: lean:au_neg_u_late_lower_pos.
au_neg_sum_late_lower_pos
Machine-readable claim: lean:au_neg_sum_late_lower_pos.
au_neg_early_inside_pos
Machine-readable claim: lean:au_neg_early_inside_pos.
finite_literal_path_decision
Machine-readable claim: lean:finite_literal_path_decision.
unit_width_path
Machine-readable claim: lean:unit_width_path.
chartAdj_iff
Machine-readable claim: lean:chartAdj_iff.
faces_connected
Machine-readable claim: lean:faces_connected.

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